

Geometry

Unit 8

Lesson 6 Homework

Name **Answer Key** _____

Date _____

Period _____

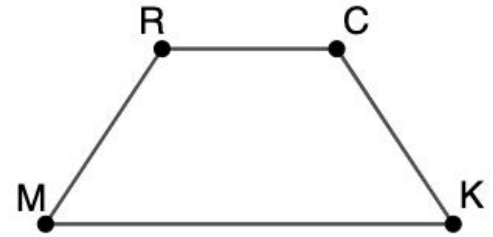
Two congruent angles isosceles trapezoid $KMRC$ have measures $(3x + 10)^\circ$ and $(5x - 10)^\circ$.

1. What is the value of x ?

$$3x + 10 = 5x - 10$$

$$20 = 2x$$

$$x = 10$$



2. What are the measures of all angles of the trapezoid?

$$m\angle M = m\angle K = 40^\circ$$

$$m\angle R = m\angle C = 140^\circ$$

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of parallel sides and diagonals are congruent.

☐ Parallelogram

☒ Rectangle

☒ Square

☐ Rhombus

☐ Kite

4. Quadrilateral $BNTV$ has diagonals are perpendicular to each other and two pairs of parallel sides.

☐ Parallelogram

☐ Rectangle

☒ Square

☒ Rhombus

☐ Kite

5. Quadrilateral $KPSX$ has two pairs of congruent sides and diagonals are perpendicular to each other.

☐ Parallelogram

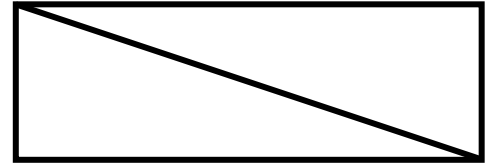
☐ Rectangle

☒ Square

☒ Rhombus

☒ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $25x$ feet and a width of $28x$ feet with a diagonal of 180 feet.



6. What is the value of x ? Round your answer to the nearest hundredth.

$$\begin{aligned}(25x)^2 + (28x)^2 &= 180^2 \\ 1409x^2 &= 32400 \\ x^2 &= 22.995 \\ x &\approx 4.80\end{aligned}$$

7. What is the height of the building? Round your answer to the nearest hundredth.

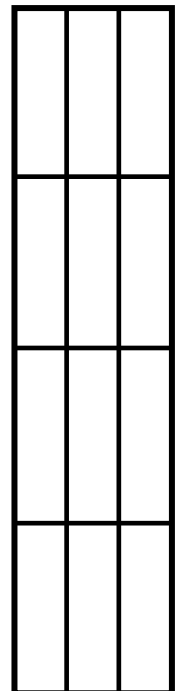
$$\text{height} \approx 25(4.80) \approx 120 \text{ feet}$$

8. What is the width of the building? Round your answer to the nearest hundredth.

$$\text{width} \approx 28(4.80) \approx 134.4 \text{ feet}$$

Use the given figure to answer questions 9-10.

A wall has paneling made up of rectangles, with an area of 30 square feet each, as shown. There are four rectangles along the height and three along the top. The edge of each rectangle in the paneling is $(2x + 3)$ feet in width and $(4x + 2)$ feet in height.



9. What is the value of x ?

$$\begin{aligned}30 &= (2x + 3)(4x + 2) \\ 0 &= 8(x + 3)(x - 1) \\ x &= -3 \text{ and } x = 1\end{aligned}$$

10. What are the dimensions of the portion of the wall covered the paneling?

$$\begin{aligned}\text{height of 1 panel} &= 4(1) + 2 = 6 \text{ feet} \\ \text{height of 4 panels} &= 4(6) = 24 \text{ feet}\end{aligned}$$

$$\begin{aligned}\text{width of 1 panel} &= 2(1) + 3 = 5 \text{ feet} \\ \text{width of 3 panels} &= 3(5) = 15 \text{ feet}\end{aligned}$$